



RTR (RADIO TELEPHONY)

QUESTION BANK

**DGCA
UPDATE 2.0**



1. Q1. How to overcome Fading?

- (A) By using AVC (Automatic Volume Control)
- (B) By using BFO (Beat Frequency Oscillator)
- (C) By Squelching



2. Q2. Which of the following equipments does not work on UHF

- (A) ILS Glidepath
- (B) ILS Localizer
- (C) DME

(Glidepath = UHF, DME = UHF)



3. Q3. If an aircrafts MCTOM is 140000kgs, in which wake turbulence category will it come ?

- (A) Medium
- (B) Heavy
- (C) Super Heavy



4. Q4. Which one of the following is not a condition that needs to be fulfilled for start up clearance?

- (A) Security Check of all persons.
- (B) Pre flight check of aircraft.
- (C) Removal of aero bridge/stairs/ramp.



FAA Aircraft Weight Class			ICAO WTC		
Code	Class	MTOW	Code	WTC	MTOM
J	Super	A380-800 and An 225	H	Heavy	≥ 136,000 kg
H	Heavy	≥ 300,000 lb (136,000 kg)	M	Medium	< 136,000 kg, > 7,000 kg
L	Large	< 300,000 lb (136,000 kg), > 41,000 lb (18,600 kg)	L	Light	≤ 7,000 kg
S	Small+	≤ 41,000 lb (18,600 kg), > 12,500 lb (5,670 kg)			
	Small	< 12,500 lb (5,670 kg)			

5. Q5. Why do we dump fuel?

- (A) Fuel quality is not good.
- (B) Fuel available is more than required.
- (C) Aircraft's weight is more than maximum landing weight.



6. Q6. Select the nationality and registration mark that can be used.

- (A) VT- TTT
- (B) VT- XXX
- (C) VT- SOS



7. Q7. If runway surface is damp it means

- (A) Surface is not affected by water, slush, snow or ice.
- (B) Significant patches of standing water are visible.
- (C) Surface shows a change of colour due to moisture.



8. Q8. Routes L, M, N, P are :

- (A) International RNAV routes.
- (B) Domestic RNAV routes.
- (C) International Non- RNAV routes.



9. Q9. Which of the following is an example of simplex mode of communication

- (A) Mobile
- (B) Fax
- (C) Landline
- (D) 10. ELT stands for



10. Q10. ELT stands for

- (A) Emergency Location Transmitter
- (B) Emergency Locator Transmitter
- (C) Emergency Local Transmitter
- (D) Emergency Location Translator



KEY TAKEAWAY

- ✓ AVC - Automatic Volume Control (fading reduction)
- ✓ BFO - Beat Frequency Oscillator
- ✓ Squelching - Reduces background noise
- ✓ Glidepath & DME work on UHF
- ✓ Wake turbulence depends on MTOW
- ✓ Start up clearance needs security & pre-flight check
- ✓ Dump fuel if landing weight is more than max allowed
- ✓ Valid registration marks follow VT- format
- ✓ Damp runway - shows change of colour
- ✓ Routes L, M, N, P are International RNAV
- ✓ Simplex - one way communication
- ✓ ELT - Emergency Locator Transmitter

REMEMBER

- ★ Glidepath = UHF
- ★ DME = UHF
- ★ Wake Turbulence → Weight Based
- ★ Start up Clearance → Security + Pre-flight
- ★ Dump Fuel → If landing weight > Max
- ★ VT-XXX → Valid Indian Registration
- ★ Damp → Change of colour
- ★ L, M, N, P → International RNAV routes
- ★ Simplex → One way
- ★ ELT → Emergency Locator Transmitter





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UPDATE 2.0**



11. Q11. Emergency phase INCERFA means

- (A) Situation wherein there is a reasonable certainty that aircraft & its occupants are threatened by grave & imminent danger and require immediate assistance.
- (B) When there is apprehend on the safety of aircraft or its occupants. Eg : aircraft fails to land within 5 mins of expected landing time.
- (C) When there is uncertainty about the safety of aircraft or its occupants.

INCERFA



12. Q12. If you exp RCF and ATC is giving light gun signals of Green Flashing light, it means

- (A) Cleared to land
- (B) Return for landing and await landing clearance
- (C) Clear to divert to alternate



AIR TRAFFIC CONTROL: LIGHT GUN SIGNALS FOR PILOTS

Aircraft on the Ground	Aircraft in Flight
Cleared for takeoff Steady green	Cleared for landing Steady green
Cleared for taxi Flashing green	Return for landing - to be followed by steady green at the proper time
STOP Steady red	Give way to other aircraft and continue circling
Taxi clear of the runway in use Flashing red	Airport unsafe, do NOT land
Return to starting point on airport Flashing white	Not applicable
Exercise extreme caution Alternating green/red	Exercise extreme caution



13. Q13. Select the correct morse code for distress

- (A) (.....-...)
- (B) (.-.-.-.-.-)
- (C) (...-.)



14. Q14. In SVFR visibility should not be less than

- (A) 1500m
- (B) 3000m
- (C) 4000m



15. Q15. The routine wx report of an aerodrome issued every half an hour is called

- (A) SIGMET
- (B) SPECI
- (C) METAR

METAR

VABB 041230Z 18010KT
6000 SCT020 28/22 Q1012
NOSIG



16. Q16. Select the correct option regarding SAR (Search & Rescue)

- (A) India is divided into 5 search and Rescue areas.
- (B) The Sub center of SAR is in Kochi.
- (C) SAR doesn't come under SARCC.



17. Q17. QTF refers to

- (A) True Bearing from the station
- (B) TAS
- (C) Position of station according to bearings taken by direction finder station.



18. Q18. Which one of the following is not a prohibited area

- (A) Tirupati Tirumala, Andhra Pradesh
- (B) Bhabha Atomic Research center
- (C) Red Fort Delhi



19. Q19. Which of the following is true regarding SSR

- (A) Not Affected by Wx.
- (B) Tx & Rx both are on ground.
- (C) Frequency is constant throughout.



KEY TAKEAWAY

- ✓ INCERFA = Uncertainty about safety of aircraft.
- ✓ Green flashing light gun = Clear to divert to alternate.
- ✓ Distress Morse Code = SOS (.....-...)
- ✓ SVFR visibility minimum = 1500m
- ✓ Routine WX report every half hour = METAR
- ✓ SAR Sub center = Kochi
- ✓ QTF = Position by DF station
- ✓ SSR is NOT affected by weather.

REMEMBER

- ★ INCERFA → Uncertainty Phase
- ★ Green Flashing → Clear to Divert
- ★ SOS → Save Our Souls
- ★ SVFR = 1500m Visibility
- ★ METAR → Regular Weather Report
- ★ SAR → Search And Rescue
- ★ QTF → Bearing to the Station
- ★ SSR → Weather Independent





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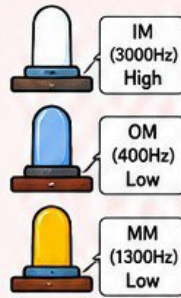
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20. Q20. Select the correct Combination of Marker Beacons

Cockpit Light	Frequency	Pitch
① IM White	3000Hz	High
② OM Blue	400Hz	Low
③ MM Amber	1300Hz	Low



21. Q21. If you are flying North of VIDP in uncontrolled airspace you will contact

- (A) ALL STATIONS on 121.5
- (B) Alpha Control
- (C) Delhi Info.



22. Q22. The phrase "I SAY AGAIN" means

- (A) I repeat for clarity
- (B) Repeat all or following part of your transmission
- (C) An error has been made in the last transmission

I SAY AGAIN

23. Q23. If we experience RCF in SID in a radar environment

- (A) We will maintain last cleared limit for 7 mins
- (B) We will maintain MEL/MEA for 7 mins
- (C) We will maintain whichever is higher between the above two for 7 mins.



24. Q24. If your TAS is 300kts what will you mention in your flight plan in item no. 15

- (A) 1<300
- (B) N300
- (C) **N0300**



25. Q25. The call sign of the ATC unit Direction Finding Station is

- (A) Finder
- (B) **Homer**
- (C) Radar



26. Q26. "IT" is the IATA code of which airline ?

- (A) **Kingfisher Airlines**
- (B) Jet Lite
- (C) Indigo Airlines



27. Q27. In Non- RVSM airspace FL390 will be

- (A) An Easterly track
- (B) A Westerly track
- (C) **Could be any of the above**



28. Q28. What does the Watch Hrs "HN" means

- (A) No specified working hours.
- (B) **Station is working sunset to sunrise.**
- (C) Station is working sunrise to sunset.



"H J" = Sunrise to sunset (Day time hrs)

29. Q29. Errors caused by the effect of Coastal refraction are maximum when the NDB is :

- (A) **Inland & the bearing crosses the coast at an acute angle.**
- (B) Near the coast and the bearing crosses the coast at right angle.
- (C) Inland & the bearing crosses the coast at right angles.



30. Q30. Decision Altitude is the Altitude

- (A) In a non-precision approach below which we can't descend.
- (B) **In a precision approach where a Missed Approach must be initiated if runway is not visible.**
- (C) Where we decide what type of approach are we going to do.



31. Q31. What is the volume 4 of Annexure 1 0

- (A) Aeronautical Charts
- (B) **Aerodrome Charts**
- (C) Helicopter Charts
- (D) Obstacle Limitation Surfaces



KEY TAKEAWAY

- ✓ Marker Beacons : IM-White (3000Hz, High), OM-Blue (400Hz, Low), MM-Amber (1300Hz, Low)
- ✓ 121.5 MHz : All Stations (Uncontrolled airspace)
- ✓ Morse Distress Code : SOS (.....)
- ✓ RCF in SID : Maintain the higher of Last Cleared Limit or MEL/MEA for 7 mins.
- ✓ TAS in FPL Item 15 : Prefix with / (e.g. 300 kts → N0300)
- ✓ DF Station Call sign : HOMER
- ✓ IT (IATA Code) : Kingfisher Airlines
- ✓ FL390 in Non-RVSM : Could be any track
- ✓ Watch Hrs "HN" : Sunset to sunrise
- ✓ Decision Altitude : MDA in Precision Approach
- ✓ Annex 10 Vol 4 : Aerodrome Charts

REMEMBER

- ★ IM - 3000Hz - High Pitch - White
- ★ OM - 400Hz - Low Pitch - Blue
- ★ MM - 1300Hz - Low Pitch - Amber
- ★ RCF - Maintain Higher (LCL / MEL/MEA) for 7 mins.
- ★ Distress Code - SOS (.....)
- ★ HN - Sunset to Sunrise
- ★ Coastal Refraction - Max error when bearing crosses coast (Acute or Right Angle).





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UPDATE 2.0**



31. Q31. What is the volume 4 of Annexure 1 0

- (A) Aeronautical Radio Frequency Spectrum Utilization.
- (B) Communication Procedures.
- (C) **Surveillance Radar & Collision Avoidance Systems.**



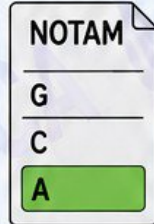
32. Q32. NOTAM Colour code yellow means

- (A) **Warning**
- (B) ATC facilities
- (C) Communication & Navigation



33. Q33. NOTAM information required by domestic flights only will come in

- (A) Series G
- (B) Series C
- (C) **Series A**



34. Q34. Annexure 15 stands for

- (A) Air Traffic Services
- (B) Aerodromes
- (C) **Aeronautical Information Services**



35. Q35. AIC (Aeronautical Information Circular) is issued by:

- (A) **DGCA**
- (B) MOCA
- (C) ICAO



36. Q36. RT phrase "RE-CLEARED" means

- (A) Permission for proposed action granted.
- (B) Authorized to proceed under conditions specified.
- (C) **A change has been made to your last clearance and this new clearance supersedes your previous clearance or part thereof.**



37. Q37. ATIS is

- (A) Automated Terminal Information Service
- (B) **Automatic Terminal Information Service**
- (C) Automatic Terminal Information System



38. Q38. VIR 156

- (A) Is a prohibited area in Delhi FIR
- (B) **Is a restricted area in Delhi FIR**
- (C) Is a restricted area in Mumbai FIR



39. Q39. If you are flying at FL350 and your Mode C fails

- (A) Will request descend to FL290
- (B) Will request descend to FL270
- (C) **Will fly offset until it becomes operational.**



40. Q40. If Hydraulic pressure is low it is

- (A) A case of RVSM suspension
- (B) A Distress Situation
- (C) **An Urgency Situation**



41. Q41. If an Aircraft has crossed at your level at about 7NM you will use the phrase

- (A) AIRPROX RISK OF COLLISION
- (B) **AIRPROX SAFETY NOT ASSURED**
- (C) AIRPROX RISK NOT DETERMINED



42. Q42. If you experience a Wind Shear on Take off path you will give

- (A) A MAYDAY call
- (B) A PAN PAN call
- (C) **A Wind Shear Report**



KEY TAKEAWAY

- ✓ Annexure 10 Vol. 4 → Surveillance Radar & Collision Avoidance Systems.
- ✓ NOTAM Yellow → Warning.
- ✓ NOTAM for Domestic Flights only → Series A.
- ✓ Annexure 15 → Aeronautical Information Services.
- ✓ AIC → Issued by DGCA.
- ✓ ATIS → Automatic Terminal Information Service.
- ✓ VIR 156 → Restricted Area in Delhi FIR.
- ✓ Hydraulic Pressure Low → Urgency Situation.
- ✓ Wind Shear on Take off → Give Wind Shear Report.

REMEMBER

- ★ AIC → Aeronautical Information Circular
- ★ RT "RE-CLEARED" → New clearance supersedes previous clearance
- ★ ATIS → Automatic Terminal Information Service
- ★ VIR 156 → Restricted Area (Delhi FIR)
- ★ RVSM → Reduced Vertical Separation Minimum
- ★ AIRPROX → Risk of collision between aircraft
- ★ Wind Shear → Report it, ensure safety





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43. Q43. If after departure your aircraft gets hit by a flock of birds resulting in structural damage

- (A) Will ask ATC to divert to nearest aerodrome
- (B) Will give a MAYDAY call
- (C) Will give a PAN PAN call



44. Q44. The IATA code of Jet Lite Airlines is

- (A) 9W
- (B) 2T
- (C) S2



45. Q45. Battery of ELT lasts for

- (A) 24hrs
- (B) 48hrs
- (C) 50hrs



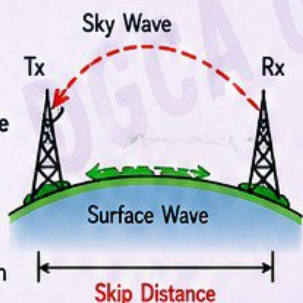
46. Q46. Which of the following is not a part of AIS (Aeronautical Information Service)

- (A) Annexures
- (B) AIRAC (Aeronautical Info Regulation & Control)
- (C) PIB (Pre flight Info. Bulletin)



47. Q47. Skip distance is

- (A) The distance between Tx & Rx.
- (B) The distance between Tx and the point on the surface where the first sky wave returns.
- (C) The area between the limit of surface wave & point of reception of first sky wave.



48. Q48. The speed of Radio wave in NM/sec is :

- (A) 300,000
- (B) 161,842
- (C) 163,842

Speed of Radio Wave
 $\approx 1.852 \times 10^5 \text{ NM/sec}$

$$\text{Speed} = \frac{186,000 \text{ miles/sec}}{60} = 161,842 \text{ NM/sec}$$

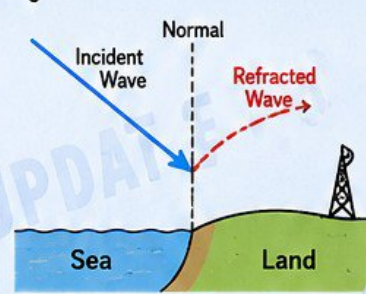
49. Q49. In which Frequency Band does Airborne Weather Radar operates

- (A) SHF
- (B) UHF
- (C) VHF



50. Q50. Radio waves traveling from sea to land

- (A) Bends towards normal
- (B) Bends away from normal
- (C) Does not bend at all



KEY TAKEAWAY

- ✓ Bird Hit with structural damage → Divert to nearest aerodrome.
- ✓ Mayday → Life threatening emergency.
- ✓ Pan Pan → Urgency situation.
- ✓ ELT Battery life → 48 hrs.
- ✓ AIS includes Annexures, AIRAC, PIB.
- ✓ Skip distance → Tx to first sky wave return point.
- ✓ Speed of Radio wave → 161,842 NM/sec.
- ✓ Airborne Weather Radar → VHF Band.
- ✓ Sea to land → Wave bends towards normal.

REMEMBER

- ★ MAYDAY → Save our Souls
- ★ PAN PAN → Urgency
- ★ ELT → Emergency Locator Transmitter
- ★ AIS → Aeronautical Information Service
- ★ VHF → Very High Frequency (30–300 MHz)
- ★ Radio Wave Speed $\approx 186,000 \text{ miles/sec}$
- ★ Refraction over Sea → Bends towards normal

